2.b.	Explain the working of a transformer with the help of a neat diagram. A 440kVA transformer is given a supply of 11kV and has 200 turns in the primary and 50 turns in the secondary. Determine: (i) Secondary induced voltage (ii) Primary full load current		06	L3	CO2
2.c.	Explain the difference between a motor and generator. For an 8-pole DC generator with 200 conductors, and flux of 30 mWb per pole, and moving at a speed of 1500 rpm, find the emf induced in the case of: (i) lap winding (ii) wave winding		06	L3	CO2
3.a.	Describe the formation and working of p-n junction diodes. Sketch and explain the forward and reverse bias characteristics of a diode.		07	L2	CO1
3.b.	Determine the value of the I_b (base current) and β in an NPN transistor, if the emitter current is 1mA and the amplification factor(α) is 0.9.		03	L3	CO3
3.c.	Explain the working of a half-wave rectifier circuit with a filter capacitor. Support with a relevant circuit diagram and waveforms		06	L2	CO3
4.a.	Explain Op-Amps with its symbol and terminals. Also, list any four ideal characteristics of an op-amp.		06	L1	CO1
4.b.	Use a summing amplifier to find the output voltage for the following sets of voltages and resistors: V_1 = 1V, V_2 = 2V, V_3 = 3V; R_1 = 500kΩ, R_2 = 1000kΩ, R_3 = 1000kΩ. Use R_f = 5MΩ.		04	L3	CO4
4.c.	Sketch the circuit configurations and write the gain formula for the following Op-Amps: (i) Integrator (ii) Differentiator		06	L2	CO4
5.a.	Using any universal gate, implement the functionality of AND, OR and NOT gates.		06	L2	CO5
5.b.	Implement the circuit connections of the JK flip-flop with NAND gates. Also, describe its functioning with the help of a truth table.		04	L2	CO5
5.c.	Sketch the block diagram of a microcontroller and describe its components.		06	L2	CO5

DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

USN: ENG25CS0453

B. Tech II Semester End Examinations – May 2026

Course Title: Single and Multivariate Calculus

Course Code: 25EN1201

Duration: 2hr 30 Min

Date: 14/May/2026

Time: 01:00 PM TO 03:30PM

Max. Marks: 80

Note:

1. **Answer all Five Full Questions**
2. **Each Full Question carries 16 Marks**
3. **Draw neat sketches wherever necessary**
4. **Missing Data may be suitably assumed**

Questions (5 × 16 = 80 Marks)

			Marks	CO	L
Q1.	a)	Show that the following function is not continuous at origin $(0,0)$ $$f(x,y)=\begin{cases}\frac{x^4-y^2}{x^4+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$	4	3	3
	b)	Find $\frac{dw}{dt}$ at $t=0$ using the Chain Rule if $w=\ln(x^2+y^2)$, $x=e^t$, and $y=\sin(t)$.	6	1	2
	c)	Use Lagrange's multiplier method to find the maximum and minimum values of $f(x,y)=2x+y$ subject to the constraint $x^2+y^2=5$.	6	4	3
Q2.	a)	Evaluate the double integral $\iint_R (2x-y)\,dy\,dx$ where R is the region bounded by $x=1, x=3, y=0$, and $y=x$.	5	2	3
	b)	Evaluate $\int_0^a\int_0^{\sqrt{a^2-x^2}}(x^2+y^2)\,dy\,dx$ by using polar coordinates.	6	2	3
	c)	Find the value of the triple integral $\int_0^1\int_0^2\int_0^{1-x-y} dz\,dy\,dx$.	5	2	4
Q3.	a)	Apply Green's Theorem to evaluate $\oint_C (xy+y^2)\,dx+x^2\,dy$ where C is the closed curve formed by $y=x$ and $y=x^2$.	5	5	3
	b)	Check whether the vector field $\vec{F}=(y^2\cos x+z^3)\hat{i}+(2y\sin x-4)\hat{j}+(3xz^2+2)\hat{k}$ is conservative. If conservative, determine the scalar potential for the given field.	6	5	3
	c)	Use Gauss Divergence Theorem to evaluate $\iint_S \vec{F}\cdot\hat{n}\,dS$, where $\vec{F}=4x\hat{i}-2y\hat{j}+z\hat{k}$ and S is the surface of the cylinder $x^2+y^2=4$ from $z=0$ to $z=3$.	5	5	3
Q4.	a)	Examine the convergence of the following series			

P.T.O